

AIR UNIT #5 — SENTINEL

ID:

obs-air-sentinel

Role: Aerial Perimeter Defense / Security Patrol

Tier: Operational

Status: Active

PHYSICAL & TECHNICAL

Length: 3.5 m

Width: 2.5 m

Height: 1.1 m

Mass: 26 kg

Power Source: High-density battery

Energy Capacity: 2300 Wh

Energy Consumption: 55 Wh/min

Endurance: 40 min

Payload Capacity: 2.5 kg

Modules: Tracking sensors

Navigation System: GPS + AI Defense Grid

Structural Integrity: 160 HP

Armor Rating: 0.25

Heat Signature: Medium

Noise Signature: Medium

Top Speed: 23 m/s

Acceleration: Medium

Turning Radius: Moderate

Climb Rate: Medium

Stability Rating: High

AUTONOMY & AI BEHAVIOR

Behavior Profile: Security / Patrol

Aggression: 0.4

Caution: 0.6

Persistence: 0.7

Flanking Likelihood: 0.2

Retreat Threshold: 0.5

Target Priority: Hostile signatures

Formation Discipline: Medium

Risk Tolerance: Medium

Sensors

Visual Range: 1500 m

Thermal Range: 900 m

Acoustic Range: Medium

Radar Range: None

LiDAR Range: None

Signal Detection Range: Medium

Communication

Mesh Range: 6 km

Relay Capacity: 0

Command Link Type: Defense Grid

Network Priority: High



COMBAT & THREAT MODEL

Firepower Rating: Low (non-lethal deterrence)

Accuracy: Medium

Stabilization: High

Weapon Mounts: 1 (non-lethal deterrent or spotlight)

Weapon Types: Tracking beam / deterrent pulse

Suppression Value: Medium

Tracking Strength: High

Disruption Strength: Medium

Breach Capability: None

Countermeasure Rating: Medium

Threat Modeling

Threat Level: 4

Engagement Distance: 200–400 m

Preferred Terrain: Perimeter airspace

Preferred Targets: Hostile drones / intruders

Vulnerability Profile: Ballistic / EMP

SQUAD & COMMAND INTEGRATION

Squad Role: Airspace Security

Squad Size Contribution: 1

Formation Behavior: Patrol loops

Intent Interpretation Rules: Maintain perimeter, intercept threats

Retreat Behavior: Medium

Breach Behavior: None

Flanking Behavior: Light

Reinforcement Behavior: Supports defense squads

Command Hierarchy

Parent Command Unit: Command Core / Sentinel (Ground)

Subordinate Units: None

Relay Nodes: None

Network Dependencies: Defense Grid

VR OPERATOR MODE

Operator Seat Type: Drone View

Operator HUD: Defense Grid Overlay

Camera Feeds: Optical

Thermal View: Yes

LiDAR View: No

Sonar View: No

Tactical Overlay: Threat markers / patrol routes

Operator Controls

Movement Controls: Full drone flight

Sensor Controls: Thermal / tracking / spotlight

Ability Controls: Mark Target, Suppress, Spotlight

Weapon Controls: Non-lethal deterrent

Stabilization Controls: Auto-hover

Operator Abilities

Threat Mark

Suppression Pulse

Spotlight Tracking

Relay Defense Data



ECONOMY & LOGISTICS

Build Cost: 200 Metal / 180 Electronics

Build Time: 60 sec

Factory Type: Drone Bay

Tech Requirements: Defense Grid I

Maintenance Cost: Medium

Repair Time: 16 sec

Supply Consumption: Medium

Operational Cost: Moderate

Deployment Requirements: Standard drone pad

Transport Compatibility: All aerial racks

Launch Method: Vertical takeoff



BUILD REQUIREMENTS (PLAYER COLLECTION SYSTEM)

Common Components

16× **Medium Alloy Plates**

12× **Composite Frame Rods**

8× **Micro-Servo Actuators**

4× **Stabilizer Fins**

1× **Tracking Sensor Array**

Electronics

14× **Circuit Bundles**

10× **Signal Processors**

6× **Drone Control Chips**

2× **Defense Grid AI Chips**

Power

1× **High-Density Battery Pack**

3× **Power Regulation Nodes**

Rare / Tradeable

1× **Threat-Detection Core**

1× **Non-Lethal Deterrent Module**

Optional Upgrades

Enhanced Spotlight

Anti-Jamming Shielding

Extended Patrol Range Module

Thermal Boost Lens